Surface Eikonal equations: Taylor–Hood formulations

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Abstract

This note aims to correct the application of formulation (5.5) in [1] to the Eikonal equation on the Möbius strip in Example 9, using the implementation in [2] as a reference. The original Taylor–Hood formulation omits an additional curvature term arising from surface integration by parts. This omission makes the formulation inconsistent with the Eikonal equation whenever the curvature coupling is nonzero. We derive the corrected weak formulation and compare the original and corrected implementations. Numerical experiments on the Möbius strip exhibit a persistent error under mesh refinement for the original formulation, providing numerical evidence that it does not converge to the exact solution.

1 Problem setting and notation

Let $\Gamma \subset \mathbb{R}^3$ be a compact, connected C^2 surface with nonempty boundary $\partial\Gamma$. We consider the Eikonal problem

$$|\nabla_\Gamma u| = 1 \quad \text{in } \Gamma \setminus \partial\Gamma, \qquad u = 0 \quad \text{on } \partial\Gamma. \tag{1}$$

The nonnegative viscosity solution is the distance function to the boundary. It is also the maximizer of

$$\max_{u\in K} \int_\Gamma u \, dS, \qquad \text{over } K = \{u \in W^{1,\infty}(\Gamma) : u|_{\partial\Gamma} = 0,\ |\nabla_\Gamma u| \le 1 \text{ a.e.}\}. \tag{2}$$

The following parts are about the notations. Choose a unit normal n locally and define

- $P = I - n \otimes n$, which is the projection operator onto the tangent space.
- $\kappa = \operatorname{div}_\Gamma n$, which is the trace of the curvature tensor.
- $\mathbf{c} = \kappa n$, which is the the curvature vector and lies on the normal direction.

Then we also need to recall the ambient derivative and surface derivative. Consider scalar function u and vector function τ. For their extensions to a neighborhood in $\mathbb{R}^3$, the ambient derivatives are difined by

$$\nabla \widetilde{u} = (\partial_j \widetilde{u})_{j=1}^3, \quad D\widetilde{\tau} = (\partial_j \widetilde{\tau}_i)_{i,j=1}^3, \quad \operatorname{div} \widetilde{\tau} = \operatorname{tr}(D\widetilde{\tau}). \tag{3}$$

Their surface counterparts are

$$\nabla_\Gamma u = P\nabla\widetilde{u}, \qquad D_\Gamma \tau = (D\widetilde{\tau})P, \qquad \operatorname{div}_\Gamma \tau = \operatorname{tr}(D_\Gamma \tau). \tag{4}$$

These surface derivatives are independent of the extensions. In contrast,

$$\nabla \widetilde{u} = \nabla_\Gamma u + (n \cdot \nabla \widetilde{u}) n \tag{5}$$

retains a normal derivative depending on the extension.

2 Surface integration by parts and corrected weak form

2.1 The integration-by-parts identity

Let $\tau : \Gamma \to \mathbb{R}^3$ be an ambient vector field. Its decomposition is

$$\tau = P\tau + (\tau \cdot n)n =: \tau_T + \tau_n n. \tag{6}$$

By chain rule, we obtain

$$\operatorname{div}_\Gamma(\tau_n n) = \nabla_\Gamma \tau_n \cdot n + \tau_n \operatorname{div}_\Gamma n = \kappa \tau_n.$$

Then the surface divergence depends on two parts,

$$\boxed{\operatorname{div}_\Gamma \tau = \operatorname{div}_\Gamma(P\tau) + \boldsymbol{\kappa(\tau \cdot n)}}. \tag{7}$$

Thus taking a surface divergence does not discard the normal component of an ambient field. Curvature couples that component into the divergence.

Apply the surface divergence theorem to the *tangent* field $u\tau_T$:

$$\int_\Gamma \operatorname{div}_\Gamma(u\tau_T)\, dS = \int_{\partial\Gamma} u\tau_T \cdot \mu\, ds. \tag{8}$$

And we can derive the integration-by-parts formula on the surface,

$$(u, \operatorname{div}_\Gamma \tau_T)_\Gamma = -(\nabla_\Gamma u, \tau)_\Gamma + \langle u, \tau \cdot \mu \rangle_{\partial\Gamma}. \tag{9}$$

Combining this with (7) yields the full identity

$$\begin{aligned}(u, \operatorname{div}_\Gamma \tau)_\Gamma &= (u, \operatorname{div}_\Gamma \tau_T)_\Gamma + (u, \kappa(\tau \cdot n))_\Gamma \\ &= -(\nabla_\Gamma u, \tau)_\Gamma + \langle u, \tau \cdot \mu \rangle_{\partial\Gamma} + \boldsymbol{(uc, \tau)_\Gamma}.\end{aligned} \tag{10}$$

For $u|_{\partial\Gamma} = 0$, the boundary term vanishes but the curvature term remains,

$$\boxed{(\nabla_\Gamma u, \tau)_\Gamma = -(u, \operatorname{div}_\Gamma \tau)_\Gamma + \boldsymbol{(uc, \tau)_\Gamma}}. \tag{11}$$

The usual identity without this term holds

- when τ is tangent, e.g. using Raviart–Thomas element.
- when the relevant curvature coupling vanishes, e.g. solving eikonal equation on the plane.

2.2 The LVPP equations and both curvature corrections

For a step $\alpha_k > 0$, initialization $\psi^0 = 0$ and using the Hellinger entropy, the gradient form of the LVPP subproblem is [1]

$$\left(\psi^k, \nabla_\Gamma v\right)_\Gamma = \alpha_k (1, v)_\Gamma + \left(\psi^{k-1}, \nabla_\Gamma v\right)_\Gamma, \qquad \forall v \in H_0^1(\Gamma). \tag{12a}$$

$$\left(\nabla_\Gamma u^k, \tau\right)_\Gamma - \left(\frac{\psi^k}{\sqrt{1 + |\psi^k|^2}}, \tau\right)_\Gamma = 0, \qquad \forall \tau \in L^2(\Gamma; \mathbb{R}^3), \tag{12b}$$

Applying the surface integration-by-parts identity to both equations gives

$$\left(\frac{\psi^k}{\sqrt{1 + |\psi^k|^2}}, \tau\right)_\Gamma + \left(u^k, \operatorname{div}_\Gamma \tau\right)_\Gamma - \boldsymbol{\left(u^k c, \tau\right)_\Gamma} = 0, \qquad \forall \tau \in H(\operatorname{div}) \tag{13a}$$

$$\left(\operatorname{div}_\Gamma(\psi^k - \psi^{k-1}), v\right)_\Gamma - \boldsymbol{\left(c \cdot (\psi^k - \psi^{k-1}), v\right)_\Gamma} + \alpha_k (1, v)_\Gamma = 0, \qquad \forall v \in L^2(\Gamma). \tag{13b}$$

Here $H(\mathrm{div})$ denotes the ambient surface-divergence space

$$H(\mathrm{div}) := \left\{\tau \in L^2(\Gamma;\mathbb{R}^3) : \mathrm{div}_\Gamma\, \tau \in L^2(\Gamma)\right\},$$

with no tangency constraint imposed on τ.

At the continuous solution, the second curvature term vanishes because ψ^k and ψ^{k-1} are tangent. But weak equations using Taylor-Hood elements do not impose exact pointwise tangency, so this term should not be dropped.

2.3 What the uncorrected ambient system expresses

If the first equation (13a) is instead

$$\left(\frac{\psi}{\sqrt{1+|\psi|^2}}, \tau\right)_\Gamma + (u, \mathrm{div}_\Gamma\, \tau)_\Gamma = 0, \tag{14}$$

then integration-by-parts (10) gives

$$\left(\frac{\psi}{\sqrt{1+|\psi|^2}} - \nabla_\Gamma u + u\mathbf{c}, \tau\right)_\Gamma = 0. \tag{15}$$

Consequently, the strong equation of (15) gives

$$\frac{\psi}{\sqrt{1+|\psi|^2}} = \nabla_\Gamma u - u\mathbf{c}, \tag{16}$$

Since $\nabla_\Gamma u$ is tangent and $u\mathbf{c}$ lies on the normal direction, this implies

$$\frac{|\psi|^2}{1+|\psi|^2} = |\nabla_\Gamma u|^2 + \kappa^2 u^2 < 1, \tag{17}$$

which is a different gradient constraint.

3 Code comparison: original and corrected formulations

3.1 Original Taylor–Hood formulation after integration by parts

The public DOLFINx implementation in [2] uses a quadratic ambient Lagrange vector field and a linear Lagrange scalar field. The comparison driver `eikonal_th_curvature.py` contains the original `TH` formulation and the explicitly corrected formulation, selected by `TH-curvature`. Both use the same element definitions:

```
el_psi = basix.ufl.element("Lagrange", cell, 2,
                           shape=(gdim,))
el_u = basix.ufl.element("Lagrange", cell, 1)
```

On the quadrilateral surface meshes, these are continuous $[Q_2]^3$ and Q_1 spaces. Neither vector space is constrained to be tangential.

For $f = \phi = 1$, the original post-integration-by-parts residual has the following implementation, with the nonlinear expression written explicitly:

```
F = (
    ufl.inner(psi / ufl.sqrt(1 + ufl.dot(psi, psi)), tau) * dx
    + u * ufl.div(tau) * dx
    + (ufl.div(psi) - ufl.div(psi_old)) * v * dx
    + self.alpha * v * dx
)
```

Here `psi_old` is ψ_h^{k-1}, and `psi` and `u` are ψ_h^k and u_h^k. The two mixed couplings are the surface divergence of the ambient test field and the surface divergence of the latent-field increment. Their curvature contributions are not subtracted in the original formulation.

3.2 Corrected Taylor–Hood formulation with explicit curvature terms

The corrected branch retains the original residual above. On each geometric cell $K \subset \Gamma_h$, it computes

$$n_h = \text{the local unit normal}, \qquad \kappa_h = \operatorname{div}_{\Gamma_h} n_h, \qquad \mathbf{c}_h = \kappa_h n_h. \tag{18}$$

The actual additional code is

```
n_h = ufl.CellNormal(mesh)
kappa_h = ufl.div(n_h)
curvature_vector = kappa_h * n_h

# F is the original divergence-form residual shown above.
if disc == "TH-curvature":
    F -= u * ufl.dot(curvature_vector, tau) * dx
    F -= v * ufl.dot(curvature_vector, psi - psi_old) * dx
```

Thus the change to the residual is precisely

$$\begin{aligned} F_{\text{corrected}} - F_{\text{original}} = & -(\boldsymbol{\kappa_h u_h^k}, \boldsymbol{\tau_h \cdot n_h})_{\boldsymbol{\Gamma_h}} \\ & -(\boldsymbol{\kappa_h(\psi_h^k - \psi_h^{k-1}) \cdot n_h}, \boldsymbol{v_h})_{\boldsymbol{\Gamma_h}}. \end{aligned} \tag{19}$$

The first term corrects the coupling between u_h^k and the divergence of τ_h. The second corrects the divergence coupling of $\psi_h^k - \psi_h^{k-1}$ with v_h. Both terms are explicitly assembled in the code and differentiated when forming the Newton Jacobian.

	Original TH	Corrected TH
Code option	`TH`	`TH-curvature`
Vector/scalar spaces	Continuous $[Q_2]^3$ / Q_1	Same spaces
Mixed couplings	Ambient surface divergences	Same divergences, with both curvature terms

Table 1: Direct comparison of the two formulations after integration by parts. The corrected branch adds the two explicit terms in (19).

Geometric implementation. The code supports degree-2 and degree-3 geometry. In both cases, $\mathbf{c}_h$ is obtained from the actual approximate surface. The coordinate element uses the same equispaced nodes as the mesh generator. Reversing a local normal changes the signs of both n_h and κ_h, leaving $\mathbf{c}_h$ and the assembled correction unchanged. Consequently a global choice of normal is not required on the Möbius strip.

4 Numerical experiments

4.1 Möbius strip, exact distance, and matched settings

We use the parametrization

$$X(\theta, s) = \big((2 + s\cos(\theta/2))\cos\theta, (2 + s\cos(\theta/2))\sin\theta, s\sin(\theta/2)\big), \tag{20}$$

where $0 \le \theta \le 2\pi$, $-1 \le s \le 1$, and periodic condition holds $(0, s) \sim (2\pi, -s)$. Since $X_\theta \cdot X_s = 0$ and $|X_s| = 1$, the exact solution is therefore

$$u^\star(X(\theta, s)) = 1 - |s|. \tag{21}$$

We compare the original formulation `TH` branch and the `TH-curvature` branch containing both explicit corrections in (19). Their continuous $[Q_2]^3/Q_1$ solution spaces, zero initialization, nonlinear solver, and step sizes are identical. The geometric coordinate degree are $g = 2$ or $g = 3$. Each mesh has $8n \times 2n$ quadrilateral cells, with $n = 2, 4, 8$; the center line $s = 0$ is aligned with mesh edges.

The reported series uses assembly quadrature degree 16 and error quadrature degree 20. Starting from $\alpha_1 = 10$, the step size doubles, with the final step clipped to reach $S = \sum_k \alpha_k = 10^4$.

4.2 Scalar errors and normal components

The two errors are

$$E_u = \|u_h - u^\star\|_{L^2(\Gamma_h)}, \qquad E_n = \left\| \frac{\psi_h}{\sqrt{1 + |\psi_h|^2}} \cdot n_h \right\|_{L^2(\Gamma_h)}. \tag{22}$$

Here

$$u^\star(x) = 1 - \sqrt{\left(\sqrt{x_1^2 + x_2^2} - 2\right)^2 + x_3^2}$$

is an ambient extension of the exact distance evaluated on Γ_h. Here E_u measures the scalar approximation error, while E_n measures the deviation of $\psi_h/\sqrt{1 + |\psi_h|^2}$ from tangency to Γ_h.

		Geometry degree $g = 2$		Geometry degree $g = 3$	
n	DOFs	Original TH	Corrected TH	Original TH	Corrected TH
2	944	5.661×10^{-2}	6.276×10^{-4}	5.653×10^{-2}	7.503×10^{-5}
4	3552	5.325×10^{-2}	7.045×10^{-5}	5.324×10^{-2}	4.690×10^{-6}
8	13760	5.148×10^{-2}	8.550×10^{-6}	5.148×10^{-2}	4.135×10^{-7}

Table 2: Scalar errors E_u. Corrected TH means the post-integration-by-parts formulation with two explicitly assembled curvature terms. DOFs count all mixed-space unknowns.

		Geometry degree $g = 2$		Geometry degree $g = 3$	
n	DOFs	Original TH	Corrected TH	Original TH	Corrected TH
2	944	1.036×10^{0}	2.426×10^{-2}	1.036×10^{0}	7.289×10^{-3}
4	3552	1.020×10^{0}	6.100×10^{-3}	1.020×10^{0}	1.807×10^{-3}
8	13760	1.013×10^{0}	1.531×10^{-3}	1.013×10^{0}	4.536×10^{-4}

Table 3: Normal-component deviation E_n for the same runs as Table 2. Both formulations use the same scalar boundary treatment.

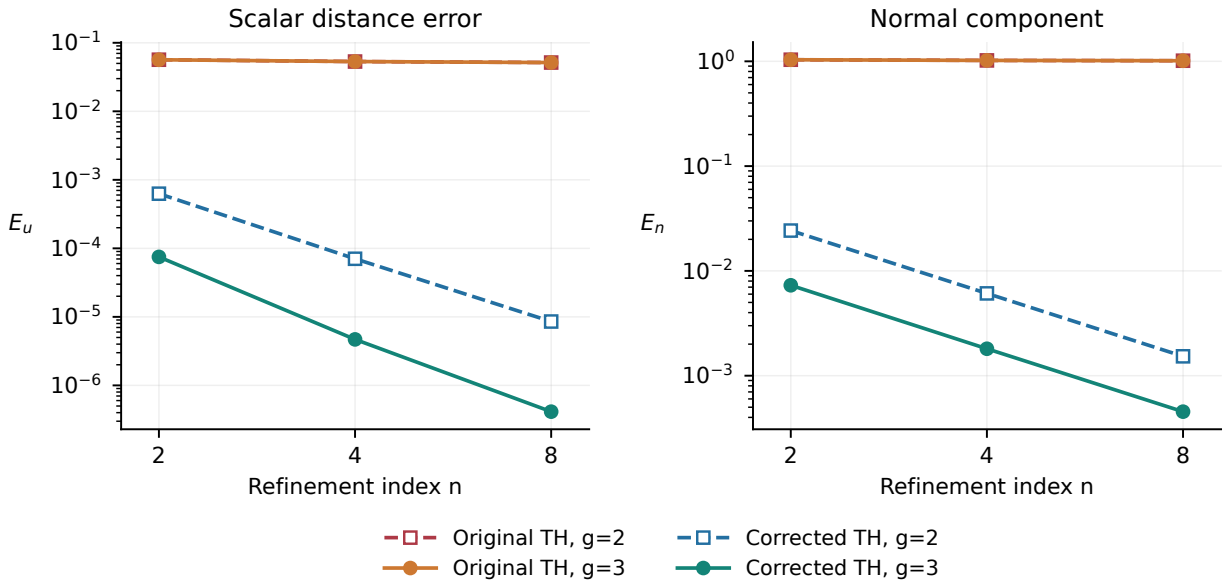


Figure 1: Original and explicitly corrected formulation on degree-2 and degree-3 geometric meshes.

For both degree-2 and degree-3 geometry, the numerical results show that the original TH formulation does not converge to the exact solution under mesh refinement, whereas the TH formulation with both explicit curvature corrections converges.

5 Code and data reproducibility

The driver `code/eikonal_th_curvature.py` implements `TH` and `TH-curvature`; the two added terms are shown in `code/curvature_terms.diff`. The reported data, source hash, and environment record are in `data/explicit_curvature/quadrature16/`. Runs used one MPI process and one BLAS/OpenMP thread with DOLFINx 0.10.0, Basix 0.10.0, UFL 2025.2.0, and PETSc 3.24.0 with MUMPS.

From the report directory, using a compatible DOLFINx Python environment with Matplotlib, run

```
EIKONAL_PYTHON=python bash code/reproduce.sh
python make_explicit_curvature_figures.py \
    --data data/reproduced/summary.json
```

The first command reproduces all twelve cases with the settings in Section 4 and writes `data/reproduced/`. The second regenerates `images/explicit_curvature_comparison.pdf` from these results.

References

[1] J. S. Dokken, P. E. Farrell, B. Keith, I. P. A. Papadopoulos, and T. M. Surowiec, *The latent variable proximal point algorithm for variational problems with inequality constraints*, Computer Methods in Applied Mechanics and Engineering **445** (2025), 118181. doi:10.1016/j.cma.2025.118181.

[2] METHODS Group, *Proximal Galerkin*, public DOLFINx Eikonal example, `examples/09_eikonal/eikonal_dolfinx.py`.

[3] FEniCS Project, UFL source documentation: gradient transformation to reference coordinates and Jacobian pseudoinverse on embedded meshes.